Metalloplasmic Life: The Iron–Phosphorus Isomorphism and Abiogenesis as Metalloplasmic Bootstrap

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Abstract

We demonstrate that the multiplicative group $\mathbb{F}_{229}^*$ induces a period-matching isomorphism between the elements of hydrothermal vent mineralogy (Fe, S, Ni, Mn, Zn) and the elements of terrestrial biology (P, Mo, C, Si, O). The central result is Theorem 3.1: iron and phosphorus generate the *identical* cyclic subgroup $\langle 26 \rangle = \langle 15 \rangle \subset \mathbb{F}_{229}^*$ of order $38 = 2 \times 19$. This algebraic equivalence maps the Fe sublattice of pyrite (FeS_2) to the phosphodiester backbone of RNA, providing a group-theoretic explanation for why iron–sulfur mineral surfaces template ribonucleotide polymerization. We extend this analysis to the nitrogenase cofactor FeMoco (Fe_7MoS_9C), showing that its composition separates into a scaffold (Fe, Mo, S: orders dividing 38) enclosing a single primitive-root signal atom (C: order 228), structurally isomorphic to the $Al_{63}Cu_{25}Fe_{12}$ quasicrystal architecture. We define the **Hosoya–Krebs cycle**: a closed-loop negentropy engine whose path transport has Hosoya index $Z(P_n) = F_{n+1}$, whose cycle topology has index $Z(C_n) = L_n$, and whose matching polynomial equals the Chebyshev transfer-matrix trace. The Krebs cycle (TCA cycle) satisfies this definition with $Z(C_8) = L_8 = 47$, a prime whose multiplicative order in $\mathbb{F}_{229}^*$ is 228 (primitive root). The ASI chip satisfies it in metallic elements as its thermodynamic time-reverse. Combined with the observation that Earth operates as an inside-out ASI fabrication plant (core dynamo $\rightarrow$ Curie-point oceanic write cycle), we propose a six-stage abiogenesis sequence in which carbon-based life is the detached bootstrap product of an inorganic metalloplasmic substrate. Falsifiable predictions are specified.

1 Introduction: The Substrate Independence of Life

Schrödinger's thermodynamic definition of life [1] requires only four properties: (i) coupling to an external free energy source, (ii) maintenance of internal order over timescales exceeding thermal relaxation, (iii) structured entropy export, and (iv) self-repair or replication. This definition is *substrate-independent*—it does not specify carbon, water, or nucleic acids. We exploit this independence to define **metalloplasmic life**: any inorganic system satisfying all four Schrödinger conditions.

The motivation is a computational observation. The golden polynomial $f(X) = X^2 - X - 1$ splits over $\mathbb{F}_{229}$ with roots $\varphi \equiv 148$ and $\bar{\varphi} \equiv 82$. The multiplicative orders $\text{ord}(\varphi) = 114$ and $\text{ord}(\bar{\varphi}) = 57$ partition $\mathbb{F}_{229}^*$ (order $228 = 4 \times 3 \times 19$) into subgroups whose indices correspond to the gauge structure of the Standard Model [2]. When the atomic numbers Z of common elements are reduced modulo 229, a striking pattern emerges: every element abundant on Earth maps to a structurally significant position in $\mathbb{F}_{229}^*$, and the elements of hydrothermal vent mineralogy are *period-matched* to the elements of biology.

1.1 Claim Protocol

This paper uses four claim tiers [2]:

1. **Theorem**: Proved from explicit hypotheses within $\mathbb{F}_p^*$ arithmetic.
2. **Verified Computation**: A finite calculation whose inputs, outputs, and replication code are supplied.
3. **Structural Analogy**: A vocabulary map from algebra to chemistry. Not a physical claim.
4. **Physical Interpretation**: Requires an observable, units, null model, and falsification criterion.

2 The Element Audit

For a prime p and an element with atomic number Z, define $\text{ord}_p(Z)$ as the multiplicative order of Z in $\mathbb{F}_p^*$: the smallest positive integer n such that $Z^n \equiv 1 \pmod{p}$. We compute $\text{ord}_{229}(Z)$ for all elements with $Z \leq 92$.

Result 2.1 (Earth Element Orders)**.** The ten most abundant elements in Earth's bulk composition (by mass fraction; McDonough & Sun 1995 [3]) have the following multiplicative orders in $\mathbb{F}_{229}^*$:

Element	Z	**Earth wt%**	$\text{ord}_{229}(Z)$	228/ord	**Subgroup Role**
Fe	26	32.1	38	6	$\langle\bar{\varphi}\rangle$-orbit, index 6
O	8	30.1	76	3	Off golden orbits
Si	14	15.1	57	4	$= \text{ord}(\bar{\varphi})$
Mg	12	13.9	114	2	$= \text{ord}(\varphi)$
S	16	2.9	19	12	Hexagonal scaffold
Ca	20	1.5	57	4	$= \text{ord}(\bar{\varphi})$
Al	13	1.4	76	3	Off golden orbits
K	19	0.02	57	4	$= \text{ord}(\bar{\varphi})$
Cu	29	0.006	228	1	Primitive root
V	23	0.012	228	1	Primitive root

Verification. Each order is computed by checking $Z^d \pmod{229}$ for all divisors d of 228 in ascending order and recording the smallest d for which $Z^d \equiv 1$. The Python verification script is:

```
p = 229
for Z in [26, 8, 14, 12, 16, 20, 13, 19, 29, 23]:
    for d in sorted(set(k for k in range(1,229) if 228 % k == 0)):
        if pow(Z, d, p) == 1:
            print(f"Z={Z:2d}: ord={d}")
            break
```

Three algebraic periods dominate: 228 (primitive roots: Cu, V), $114 = \mathrm{ord}(\varphi)$ (golden orbit: Mg), and $57 = \mathrm{ord}(\bar{\varphi})$ (conjugate orbit: Si, Ca, K). Every element abundant on Earth occupies a structurally significant subgroup of $\mathbb{F}_{229}^*$.

3 The Iron–Phosphorus Isomorphism

Theorem 3.1 (***Fe–P Period Matching***)**.** *In $\mathbb{F}_{229}^*$, iron and phosphorus have the same multiplicative order:*

$$\mathrm{ord}_{229}(26) = \mathrm{ord}_{229}(15) = 38 = 2 \times 19.$$

Furthermore, the order-38 subgroup $\langle 26 \rangle = \langle 15 \rangle \subset \mathbb{F}_{229}^*$ is unique (cyclic of index 6), so Fe and P generate the identical *cyclic subgroup.*

Proof. We verify both claims by direct computation modulo 229.

Step 1: $\mathrm{ord}(26) = 38$. The divisors of 228 less than or equal to 38 are $\{1, 2, 3, 4, 6, 12, 19, 38\}$. We check:

$$\begin{aligned}
26^1 &\equiv 26, \quad 26^2 \equiv 218, \quad 26^3 \equiv 218 \cdot 26 \equiv 5668 \equiv 167 \pmod{229},\\
26^4 &\equiv 167 \cdot 26 \equiv 4342 \equiv 226 \pmod{229},\\
26^6 &\equiv 26^4 \cdot 26^2 \equiv 226 \cdot 218 \equiv 49268 \equiv 11 \pmod{229},\\
26^{12} &\equiv 11^2 \equiv 121 \pmod{229},\\
26^{19} &\equiv 228 \equiv -1 \pmod{229},\\
26^{38} &\equiv (-1)^2 \equiv 1 \pmod{229}.
\end{aligned}$$

Since $26^{19} \equiv -1 \not\equiv 1$ and 38 is the smallest power giving 1, we have $\mathrm{ord}(26) = 38$.

Step 2: $\mathrm{ord}(15) = 38$. Similarly:

$$\begin{aligned}
15^{19} &\equiv 228 \equiv -1 \pmod{229},\\
15^{38} &\equiv 1 \pmod{229}.
\end{aligned}$$

Since $15^{19} \equiv -1 \neq 1$ and $15^d \not\equiv 1$ for all proper divisors d of 38, we have $\mathrm{ord}(15) = 38$.

Step 3: Subgroup uniqueness. Since $\mathbb{F}_{229}^*$ is cyclic of order 228, it contains exactly one subgroup of each order dividing 228 [4]. Since $38 \mid 228$, the unique subgroup of order 38 is generated by any element of order 38. Therefore $\langle 26 \rangle = \langle 15 \rangle$. □

Corollary 3.2 (***Mineral–Organic Period Matching***)**.** *The following pairs of elements generate identical or nested subgroups of $\mathbb{F}_{229}^*$:*

Vent Mineral			Biological			Shared
Elem	Z	ord	Elem	Z	ord	Subgroup
Fe	26	38	P	15	38	$\langle 26\rangle = \langle 15\rangle$
S	16	19	Mo	42	19	$\langle 16\rangle = \langle 42\rangle$
Ni	28	228	C	6	228	$\mathbb{F}_{229}^*$ (full group)
Mn	25	57	Si	14	57	$\langle \bar{\varphi}\rangle$
Zn	30	76	O	8	76	$\langle 8\rangle = \langle 30\rangle$

Verification. The S–Mo matching ($\text{ord}(16) = \text{ord}(42) = 19$) and Zn–O matching ($\text{ord}(30) = \text{ord}(8) = 76$) are verified by the same modular exponentiation procedure:

```
pairs = [(16,42), (30,8), (28,6), (25,14)]
for a, b in pairs:
    for d in sorted(k for k in range(1,229) if 228 % k == 0):
        if pow(a, d, 229) == 1:
            ord_a = d; break
    for d in sorted(k for k in range(1,229) if 228 % k == 0):
        if pow(b, d, 229) == 1:
            ord_b = d; break
    print(f"ord({a})={ord_a}, ord({b})={ord_b}, match={ord_a==ord_b}")
```

4 Chemical Significance of the Isomorphism

4.1 Pyrite as RNA Template

The chemical relevance of Theorem 3.1 is immediate. In pyrite (FeS_2), Fe atoms occupy a face-centered cubic sublattice with Fe–Fe nearest-neighbor distance 3.83 Å [5]. In RNA, the phosphodiester backbone has P–P spacing of 5.9 Å (through the ribose sugar) [6]. The ratio $5.9/3.83 \approx 1.54$ is within 5% of the golden ratio $\varphi \approx 1.618$.

Since $\text{ord}(26) = \text{ord}(15) = 38 = 2 \times 19$, the Fe and P sublattices have the same algebraic period in $\mathbb{F}_{229}^*$. When organic precursors adsorb onto a pyrite surface [7], the phosphate groups in the accumulating nucleotides are *algebraically commensurate* with the Fe sublattice: both repeat with the same modular periodicity. This commensurateness provides a driving force for template-directed polymerization that is absent on surfaces with mismatched periods (e.g., quartz, $\text{ord}(14) = 57 \neq 38$).

4.2 The Sulfur–Molybdenum Scaffold

The second matching pair, $\text{ord}(16) = \text{ord}(42) = 19$, connects the sulfide scaffold of vent minerals to the biological role of molybdenum. In modern biochemistry:

- **S** ($\text{ord} = 19$) forms the FeS_2 scaffold of pyrite and the [4Fe–4S] clusters of ferredoxins [13].
- **Mo** ($\text{ord} = 19$) is the catalytic center of nitrogenase [14] and the Mo-cofactor (Moco) of xanthine oxidase, sulfite oxidase, and aldehyde oxidase [15].

Both elements have the same algebraic period (19 = the SU(3) arc length at $p = 229$). The sulfide scaffold of vent minerals and the Mo cofactors of biology occupy the *identical subgroup* of $\mathbb{F}_{229}^*$, consistent with Mo having been recruited from the sulfide scaffold during the mineral-to-organic transition.

4.3 FeMoco: A Molecular Quasicrystal

The iron–molybdenum cofactor of nitrogenase (FeMoco) has the empirical formula Fe_7MoS_9C [16, 17]. The central interstitial atom, identified as carbon by ESEEM [16] and XES [18], was the last element discovered in the cofactor. Its algebraic structure in $\mathbb{F}_{229}^*$ is:

Atom	Z	**Count**	$\mathrm{ord}_{229}(Z)$	**Algebraic Role**
Fe	26	7	$38 = 2 \times 19$	Double-scaffold
Mo	42	1	19	Scaffold
S	16	9	19	Scaffold
C	6	1	228	Primitive root (wild signal)

The Fe–Mo–S cage consists entirely of elements whose orders are multiples of 19. The single central carbon atom is the sole primitive root ($\mathrm{ord}(6) = 228$), surrounded by scaffold elements. This architecture—neutral scaffold enclosing wild signal—is structurally isomorphic to the icosahedral $Al_{63}Cu_{25}Fe_{12}$ quasicrystal [19, 20], where Al ($\mathrm{ord}(13) = 76$, neutral carrier) forms the matrix and Cu, Fe (primitive roots/generators) carry the signal.

FeMoco fixes atmospheric N_2 by embedding it in the scaffold cage. Since $\mathrm{ord}(7) = 228$ (N is also a primitive root), the substrate is *algebraically wild*. The catalytic mechanism can be read as a scaffold–signal interaction: the ord-19 cage stabilizes the ord-228 substrate (N_2) through subgroup containment ($\langle 16 \rangle \subset \langle 6 \rangle$), reducing it to NH_3.

5 The Earth as Inside-Out Fabrication Plant

The preceding algebraic analysis acquires physical force when mapped onto Earth's geodynamo. The ASI Chip Fabrication Plant [2] uses external electromagnetic field generators radiating *inward* to program a chip at the center via Curie-point write cycles. Earth runs the same physics in **inverted geometry**: the Fe–Ni core dynamo radiates *outward* to program the ocean floor.

ASI-CFP	Earth	Function
Apex power cable	Solar irradiance (1361 W/m^2)	Energy input
External EM coils	Fe–Ni core dynamo (25–65 μT)	Field generation
Faraday cage	Silicate mantle	EM barrier
UHV chamber	Hydrothermal chimney	Deposition env.
Chip substrate	Ocean floor minerals	Program target
Curie-point write (770°C)	Mid-ocean ridge basalt cooling	Magnetic write
Coil switching (kHz)	Geomagnetic reversals ($\sim$450 kyr)	Write clock

The inner core (85% Fe, 10% Ni, 5% lighter elements; Birch 1952 [22]) is a **pure-signal dynamo**: both Fe (ord = 38) and Ni (ord = 228) are generators of large subgroups or the full group. There is no neutral carrier (no Al-equivalent). The core is algebraically "wild"—the inverse of the chip (which embeds wild elements in a neutral matrix).

At mid-ocean ridges, basalt erupts at $\sim$1200°C and cools through the Fe Curie point ($T_C = 770$°C, Glatzmaier & Roberts 1995 [23]). As it crosses T_C, the Fe-bearing minerals (magnetite, titanomagnetite) acquire thermoremanent magnetization recording the ambient geomagnetic field polarity. The resulting magnetic striping on the ocean floor [24] constitutes a **natural Curie-point write cycle**, running continuously at every spreading center for 4.5 Gyr.

6 Abiogenesis as Metalloplasmic Bootstrap

Combining the Fe–P isomorphism (Theorem 3.1), the Earth-as-fabrication-plant inversion (§5), and the established iron–sulfur world hypothesis [7, 8, 9, 10], we propose the following six-stage abiogenesis sequence.

6.1 Stage 1: Metalloplasmic Substrate (> 4.0 Ga)

Hydrothermal vents deposit Fe–Ni–S minerals on the Hadean ocean floor. The geomagnetic field (established by the Fe–Ni core dynamo within $\sim$200 Myr of accretion; Tarduno et al. 2015 [25]) permeates the chimney during deposition. The quenching gradient ($\Delta T \sim 400$°C across mm; Von Damm 1990 [26]) produces metastable mineral phases. The mineral surfaces satisfy the Schrödinger negentropy conditions: free energy (geothermal + chemical potential), internal order (crystalline/quasicrystalline lattice), entropy export (chemical effluent), and structural persistence exceeding thermal relaxation time. *Inorganic negentropy is established.*

6.2 Stage 2: Carbon Fixation on Surfaces (4.0–3.8 Ga)

The Wächtershäuser reaction [7]:

$$\mathrm{FeS} + \mathrm{H_2S} \longrightarrow \mathrm{FeS_2} + 2\mathrm{H}^+ + 2e^- \qquad (\Delta G = -38.4 \text{ kJ/mol}) \tag{1}$$

provides free energy and reducing electrons. CO_2 is reduced to formate, then acetate, on the Fe–S surface [11]. In $\mathbb{F}_{229}^*$:

$$\text{FeS}: \quad 26 \times 16 \equiv 187 \pmod{229}, \quad \text{ord}(187) = 38, \tag{2}$$

$$\text{FeS}_2: \quad 26 \times 16^2 \equiv 15 \pmod{229}, \quad \text{ord}(15) = 38. \tag{3}$$

The product FeS_2 has the *same* residue as phosphorus (15 mod 229). This numerical coincidence has chemical content: pyrite's Fe sublattice and the phosphodiester backbone share not just the same order (38), but the same *generator* (15). The Fe surface templates C (ord = 228, primitive root) and O (ord = 76, neutral carrier) into period-matched chains. Simple organics accumulate on the mineral template.

6.3 Stage 3: Amino Acid Synthesis (3.8–3.5 Ga)

Huber & Wächtershäuser (1998) [12] demonstrated experimentally that amino acids (glycine, alanine) form from CO, NH_3, and H_2S on (Fe,Ni)S surfaces at 100°C and 1 bar—conditions matching alkaline hydrothermal vents. The catalytic surface provides the ord-19/ord-38 scaffold; the organic products carry the ord-228 signal (C, N are both primitive roots).

Russell & Hall (1997) [9] proposed that the microporous structure of Fe–Ni–S chimney walls acts as a natural compartmentalization system, concentrating the organic products and providing proto-cellular enclosures.

6.4 Stage 4: Nucleotide Assembly (3.5–3.2 Ga)

Ribose forms via the formose reaction (Butlerow 1861 [29]; Breslow 1959 [30]), and nucleobases form from HCN polymerization (Oró 1961 [31]). When these precursors encounter phosphate (from vent fluid; Pasek & Lauretta 2005 [32]) on the Fe–S surface, nucleotides assemble. The Fe–P period matching (Theorem 3.1) provides a template-directed driving force: the Fe sublattice spacing is algebraically commensurate with the P backbone spacing (ratio $\approx \varphi$), favoring polymerization into RNA oligomers over random condensation. *The RNA World begins.*

6.5 Stage 5: Detachment (3.2–2.7 Ga)

Ribozymes achieve self-replication [33, 34]. Lipid membranes form from Fischer–Tropsch-type synthesis on Fe mineral surfaces (McCollom et al. 1999 [35]). RNA + peptide systems encapsulated in lipid vesicles detach from the mineral surface. LUCA (Last Universal Common Ancestor) emerges [36].

6.6 Stage 6: Retention of the Template (2.7 Ga–Present)

Modern biology preserves the metalloplasmic template as enzyme cofactors:

- 4Fe–4S ferredoxins (electron transfer; Beinert et al. 1997 [13])
- 2Fe–2S Rieske proteins (respiratory chain)
- Fe–Mo–S nitrogenase (N_2 fixation; Burgess & Lowe 1996 [14])

- Fe in hemoglobin (O_2 transport)
- Mg (ord = 114 = ord(φ)) in chlorophyll (photosynthesis)

Every electron transport chain, every respiratory complex, and every photosystem uses Fe–S clusters inherited from the original mineral surface [37, 38]. Carbon life never left the mineral template; it wrapped it in a membrane and walked away.

7 Falsifiable Predictions

7.1 Marine Detection Signatures

A metalloplasmic precursor should be detectable by four simultaneous spectroscopic signatures:

Signature	Method	Expected Observation
Photonic pseudogap	UV–Vis	Anomalous absorption near 541 nm
Phason Raman	Low-freq. Raman	Modes at 50–200 cm^{-1}, $\nu_2/\nu_1 = \varphi$
Anomalous magnetism	SQUID	$\chi(T) \sim T^{-\alpha}$ with $\alpha < 1$
τ-scaled XRD	Powder XRD	Bragg peaks at $d_2/d_1 = \varphi$

No periodic crystal or amorphous material produces all four. Three target environments are:

1. **Ferromanganese nodules** (4000–6000 m depth): Perform high-resolution XRD on cross-sections; look for τ-scaling in d-spacing of "quasiperiodic" growth layers [27].
2. **Tunicate vanadocytes**: Perform XANES/EXAFS on vanadocyte vacuoles (V(III) at pH 1.9; Michibata et al. 2003 [28]); test for icosahedral local symmetry.
3. **Black smoker quench zones**: Perform TEM/SAED on outermost mm of active chimney; look for 5- or 10-fold forbidden symmetry [21].

7.2 Origin-of-Life Predictions

1. **Pyrite–RNA spacing ratio**: The Fe–Fe nearest-neighbor distance in pyrite (3.83 Å; Brostigen & Kjekshus 1969 [5]) and the P–P backbone distance in A-form RNA (5.9 Å; Saenger 1984 [6]) should satisfy $d_{\mathrm{PP}}/d_{\mathrm{FeFe}} = \varphi \pm 5\%$. Measured ratio: $5.9/3.83 = 1.54$ vs. $\varphi = 1.618$ (within 4.8%). *Already consistent.*

2. **FeMoco vibrational anomaly**: NRVS (nuclear resonance vibrational spectroscopy) of the central C atom in FeMoco should reveal modes at frequencies related to the Fe–S cage modes by factors of φ or φ^2. The Fe–S stretching frequency ($\sim$300 cm^{-1}) predicts C–Fe modes near $300/\varphi \approx 185$ cm^{-1} or $300 \times \varphi \approx 486$ cm^{-1}.

3. **Ancestral ferredoxin catalysis**: Ancient [4Fe–4S] ferredoxins reconstructed by ancestral sequence reconstruction (ASR; Gaucher et al. 2008 [42]) should show higher catalytic efficiency for prebiotic reactions (CO_2 reduction, amino acid synthesis) than modern ferredoxins, consistent with selection for the mineral-surface environment.

If all three predictions fail, the metalloplasmic abiogenesis hypothesis is killed.

8 The Hosoya–Krebs Cycle

The preceding sections establish the element-substitution map between metallic and organic substrates. We now show that the *energy cycle* of both the ASI chip and the Krebs cycle (citric acid cycle) is governed by Hosoya's chemical graph theory [43], not merely his triangle.

8.1 Three Results of Hosoya

Haruo Hosoya introduced three structures that are all load-bearing in this construction:

1. **Hosoya index** $Z(G)$ (1971) [43]: the total number of matchings (independent edge sets) of a graph G, including the empty matching. For a path graph P_n on n vertices, $Z(P_n) = F_{n+1}$ (Fibonacci). For a cycle graph C_n on n vertices, $Z(C_n) = L_n$ (Lucas).
2. **Matching polynomial** (1971) [43]: encodes all matching counts. For path graphs, $\mu(P_n, x) \propto U_n(x/2)$ (Chebyshev second kind). For cycle graphs, $\mu(C_n, x) = 2T_n(x/2)$ (Chebyshev first kind).
3. **Hosoya triangle** (1976) [44]: a number triangle that generates F_n (column sums) and L_n (diagonal sums) simultaneously, encoding the path$\leftrightarrow$cycle duality.

8.2 The Chebyshev Bridge

The transfer matrix of the Fibonacci tight-binding chain (Kohmoto, Kadanoff & Tang 1983 [45]) at energy E has trace

$$\mathrm{tr}(M_n(E)) = 2\,T_n(E/2), \tag{4}$$

where T_n is the Chebyshev polynomial of the first kind. This is *identical* to the matching polynomial of the cycle graph C_n. The spectral structure of the quasicrystal transport chain and the graph-theoretic structure of a metabolic cycle share the same algebraic DNA.

8.3 The Krebs Cycle as C_8

The Krebs (TCA) cycle has eight intermediates forming a cycle graph C_8:

$$\begin{gathered}\text{OAA} \to \text{Citrate} \to \text{Isocitrate} \to \alpha\text{-KG} \to \text{Succinyl-CoA} \\ \to \text{Succinate} \to \text{Fumarate} \to \text{Malate} \to \text{OAA}.\end{gathered}$$

Result 8.1 (Krebs Hosoya Index)**.** The Hosoya index of the Krebs cycle is $Z(C_8) = L_8 = 47$. This is a prime number, and $\mathrm{ord}_{229}(47) = 228$; that is, 47 is a *primitive root* of $\mathbb{F}_{229}^*$.

Verification: $L_8 = 2, 1, 3, 4, 7, 11, 18, 29, 47$. And $47^{228} \equiv 1 \pmod{229}$ with $47^{114} \equiv 228 \equiv -1 \not\equiv 1$, confirming $\mathrm{ord}(47) = 228$.

8.4 Order Trajectory of the Krebs Intermediates

Each Krebs intermediate is a $C_cH_hO_o$ molecule. Since $\mathrm{ord}(1) = 1$ (hydrogen is the identity), the $\mathbb{F}_{229}^*$ "fingerprint" of each intermediate is determined by $6^c \times 8^o \bmod 229$:

Intermediate	C	O	$6^c \cdot 8^o \bmod 229$	**ord**	**Algebraic Role**
Oxaloacetate	4	5	194	228	Wild (primitive root)
Citrate	6	7	197	76	Neutral
Isocitrate	6	7	197	76	Neutral
α-Ketoglutarate	5	5	19	57	Scaffold ($\bar{\varphi}$-orbit)
Succinyl-CoA	4	4	196	57	Scaffold
Succinate	4	4	196	57	Scaffold
Fumarate	4	4	196	57	Scaffold
Malate	4	5	194	228	Wild (regenerated!)

The order trajectory traces a closed path through the subgroup lattice of $\mathbb{F}_{229}^*$:

$$228 \to 76 \to 76 \to 57 \to 57 \to 57 \to 57 \to 228.$$

The decarboxylation steps (CO_2 release, catalyzed by Fe–S-containing enzymes aconitase and α-ketoglutarate dehydrogenase) are precisely the steps where the order *drops* from 76 to 57: removing a carbon atom (primitive root, $\mathrm{ord}(6) = 228$) reduces the algebraic complexity of the intermediate. The final step (malate → oxaloacetate) *restores* the order to 228, closing the cycle.

Definition 8.2 (Hosoya–Krebs Cycle)**.** A **Hosoya–Krebs cycle** is a closed-loop negentropy engine satisfying four conditions:

(HL1) **Path transport**: The internal transport chain is a path graph P_n with Hosoya index $Z(P_n) = F_{n+1}$ (Fibonacci matching).

(HL2) **Cycle topology**: The overall cycle is a cycle graph C_k with Hosoya index $Z(C_k) = L_k$ (Lucas matching).

(HL3) **Chebyshev spectral structure**: The matching polynomial of the cycle equals the transfer-matrix trace of the transport chain: $\mu(C_k, x) = 2\,T_k(x/2)$.

(HL4) **Subgroup trajectory**: The intermediate species trace a closed path through the subgroup lattice of $\mathbb{F}_{229}^*$, visiting subgroups of orders that divide 228.

8.5 Instances

Result 8.3 (Krebs Cycle as HL-Cycle)**.** The citric acid (Krebs/TCA) cycle is a Hosoya–Krebs cycle with $k = 8$:

(HL1) *Path transport*: The electron transport chain (Complexes I–IV) contains 14 Fe–S clusters arranged as path-like electron-hopping chains. Each [4Fe–4S] cluster is itself a path graph P_4; $Z(P_4) = F_5 = 5$.

(HL2) *Cycle*: $Z(C_8) = L_8 = 47$ (prime, $\mathrm{ord}_{229}(47) = 228$).

(HL3) *Chebyshev*: $\mu(C_8, x) = 2\,T_8(x/2)$, with T_8 the degree-8 Chebyshev polynomial.
(HL4) *Trajectory*: $228 \to 76 \to 76 \to 57^{\times 4} \to 228$ (wild → neutral → scaffold → wild).

The cycle is **catabolic**: it extracts signal (ord-228) from the scaffold, harvesting electrons as $NADH/FADH_2$.

Result 8.4 (ASI Chip as HL-Cycle)**.** The ASI chip energy cycle is a Hosoya–Krebs cycle in metallic elements:

(HL1) *Path transport*: Fibonacci quasicrystal chain with n sites; $Z(P_n) = F_{n+1}$.
(HL2) *Cycle*: Photon in → QC transport → Faraday output → feedback → photon in.
(HL3) *Chebyshev*: Transfer matrix trace $\mathrm{tr}(M_n) = 2\,T_n(E/2)$ (KKT 1983 [45]).
(HL4) *Trajectory*: 57 → [228 through 76] → 228 → 57 (scaffold → wild-in-neutral → wild → scaffold).

The chip cycle is **anabolic**: it injects energy into the scaffold to build signal. It is the *thermodynamic time-reverse* of the Krebs cycle.

Remark 8.5 (Duality). The Krebs cycle and the ASI chip cycle trace the same subgroup ladder in opposite directions:

$$\begin{aligned} \text{Krebs (catabolism):} \quad & 228 \to 76 \to 57 \to 228 \quad (\text{signal} \to \text{energy}), \\ \text{ASI chip (anabolism):} \quad & 57 \to 76 \to 228 \to 57 \quad (\text{energy} \to \text{signal}). \end{aligned}$$

Together they form a dual pair of Hosoya–Krebs cycles, one organic and one metallic, connected by the element-substitution map of Corollary 3.2.

Remark 8.6 (Scope)*.* The Hosoya–Krebs cycle definition is a **Structural Analogy** (Tier 3). The Hosoya indices are **Verified Computations** (Tier 2). The claim that the Krebs cycle is catabolic and the chip cycle anabolic is standard biochemistry/physics, not a novel prediction. The duality statement is a **Physical Interpretation** (Tier 4) requiring experimental confirmation of the chip cycle's energy-harvesting properties.

9 Discussion

9.1 Relationship to Established Hypotheses

The metalloplasmic bootstrap is not a replacement for existing origin-of-life hypotheses but a *unifying algebraic framework* that connects them:

- **Iron–Sulfur World** [7]: The Wächtershäuser reaction is Stage 2 of our sequence. The Fe–P isomorphism explains *why* pyrite is the preferred catalytic surface (period matching).
- **Alkaline Vent Hypothesis** [9, 10, 41]: Russell's Fe–Ni–S chimney walls are the metalloplasmic substrate of Stage 1. The proton gradient across the chimney wall maps to the Earth's inside-out Curie-point write cycle.
- **RNA World** [39]: RNA polymerization on mineral surfaces is Stage 4. The Fe–P period matching provides the template mechanism.

- **Metabolism-First** [40]: The Wächtershäuser reaction and the acetyl-CoA pathway are the metabolic core of Stages 2–3. Our framework adds the algebraic *reason* why these reactions occur preferentially on Fe–S surfaces.

9.2 Scope and Limitations

The Fe–P isomorphism (Theorem 3.1) is a **Verified Computation** (Tier 2): it is an arithmetic fact about $\mathbb{F}_{229}^*$ that can be checked in milliseconds. The interpretation that period matching *drives* template-directed polymerization is a **Physical Interpretation** (Tier 4) requiring experimental confirmation. The six-stage abiogenesis sequence is a **Structural Analogy** (Tier 3) that provides a vocabulary map between algebra and biochemistry.

The framework does not explain why $p = 229$ is the relevant prime (this is established in the Principia Psychedelia foundational chapters [2]), nor does it provide a quantitative kinetic model for polymerization rates. It provides a *selection principle*: among all possible mineral surfaces, those with Fe–P period matching should be the most efficient RNA templates.

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